Scanline

Production matchers often sweep a scanline across edges instead of $O(n^2)$ pairs only

The idea

- M1 has 3 shapes → 6 events (enter/leave each)
- Events sort by x then enter-before-leave

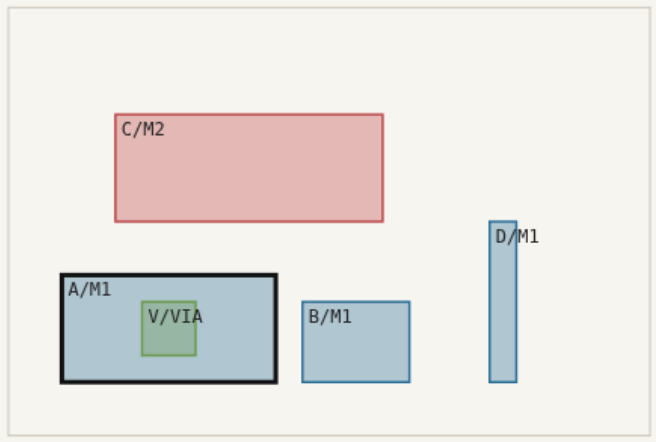
Enter

SCANLINE SWEEP

Enter

Step 1 / 5 · enter · module02-01-scanline-sweep

enter x0



Explanation

Enter event at shape x0.

- enter

```
x0
```

Enter

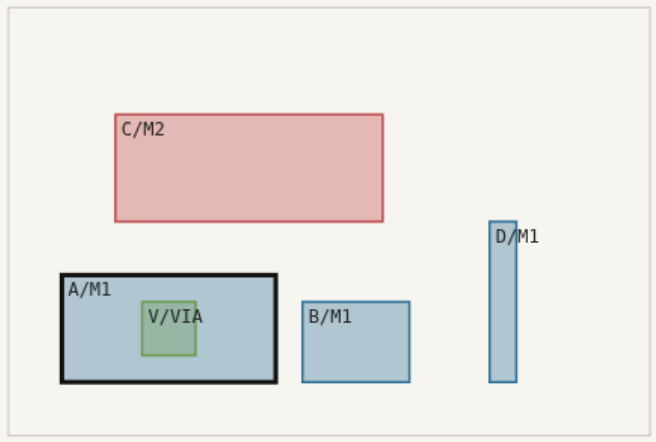
Leave

SCANLINE SWEEP

Leave

Step 2 / 5 · leave · module02-01-scanline-sweep

leave x1



Explanation

Leave event at shape x1.

- leave

```
x1
```

Leave

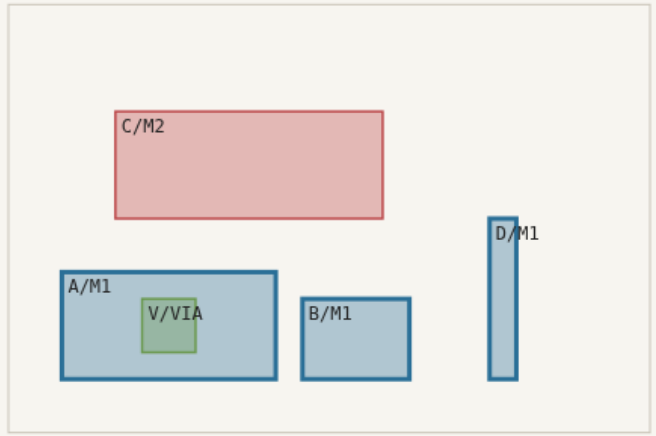
Sort

SCANLINE SWEEP

Sort

Step 3 / 5 · sort · module02-01-scanline-sweep

sorted



Explanation

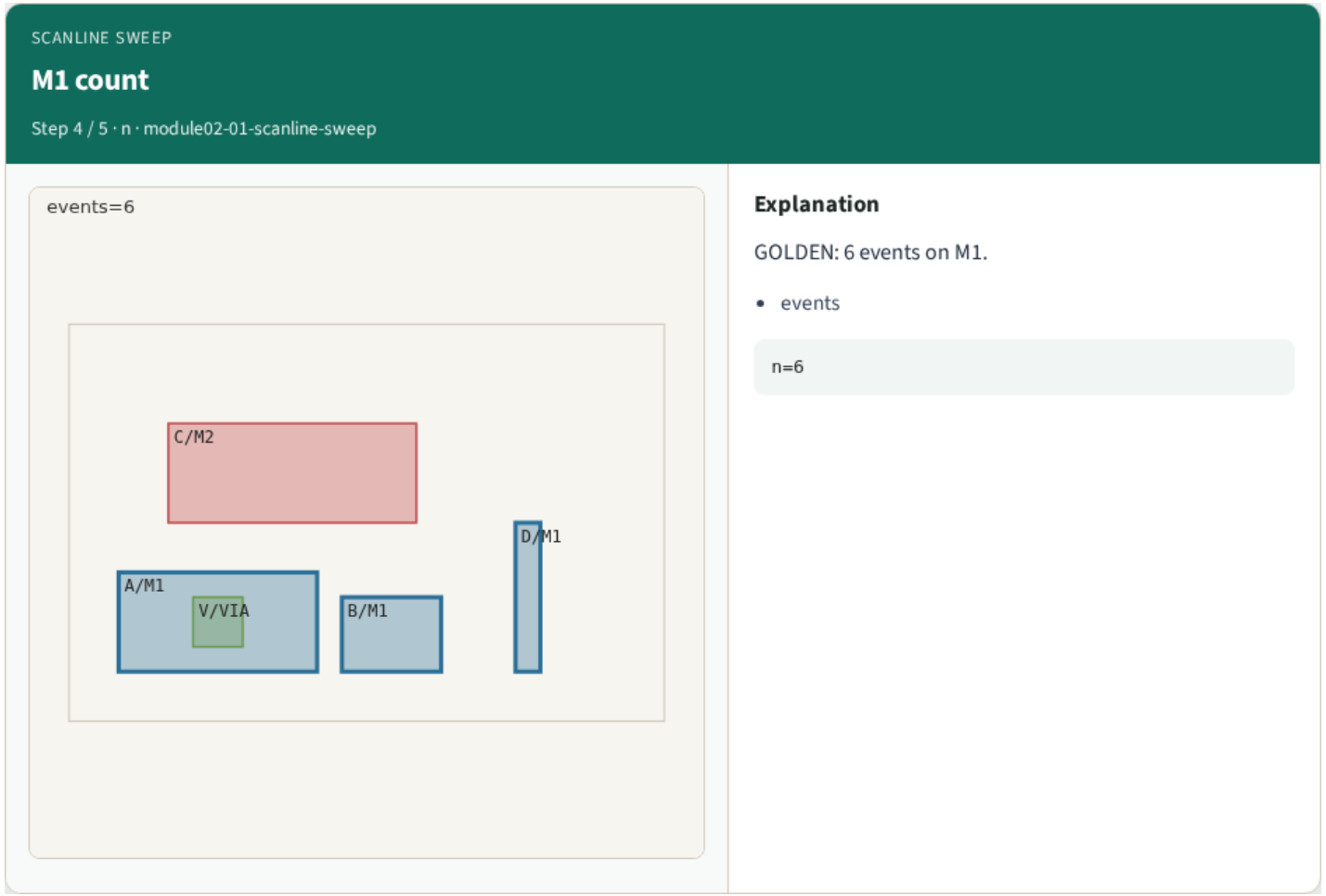
Sort by x; enters before leaves at same x.

- order

```
sorted
```

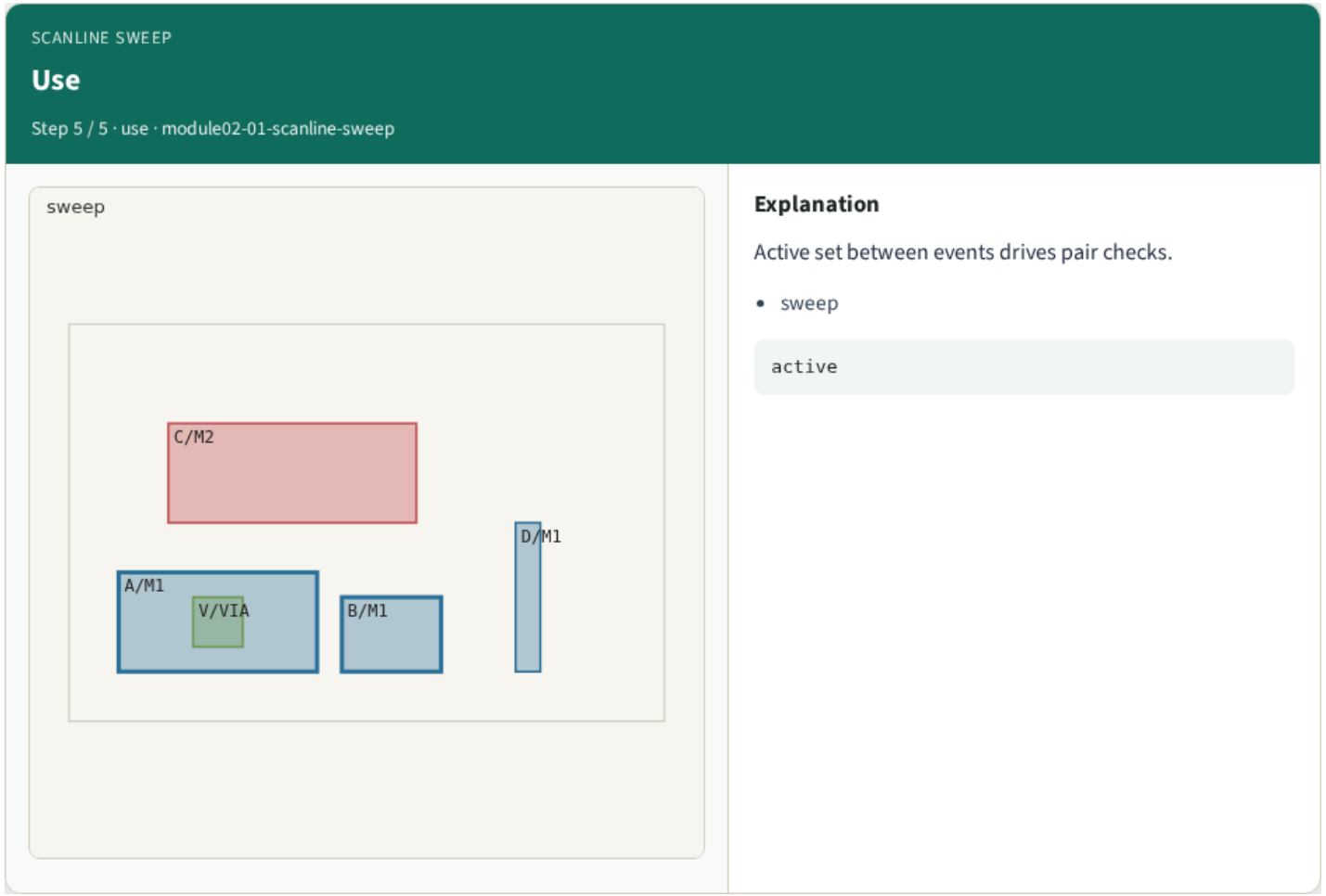
Sort

M1 count



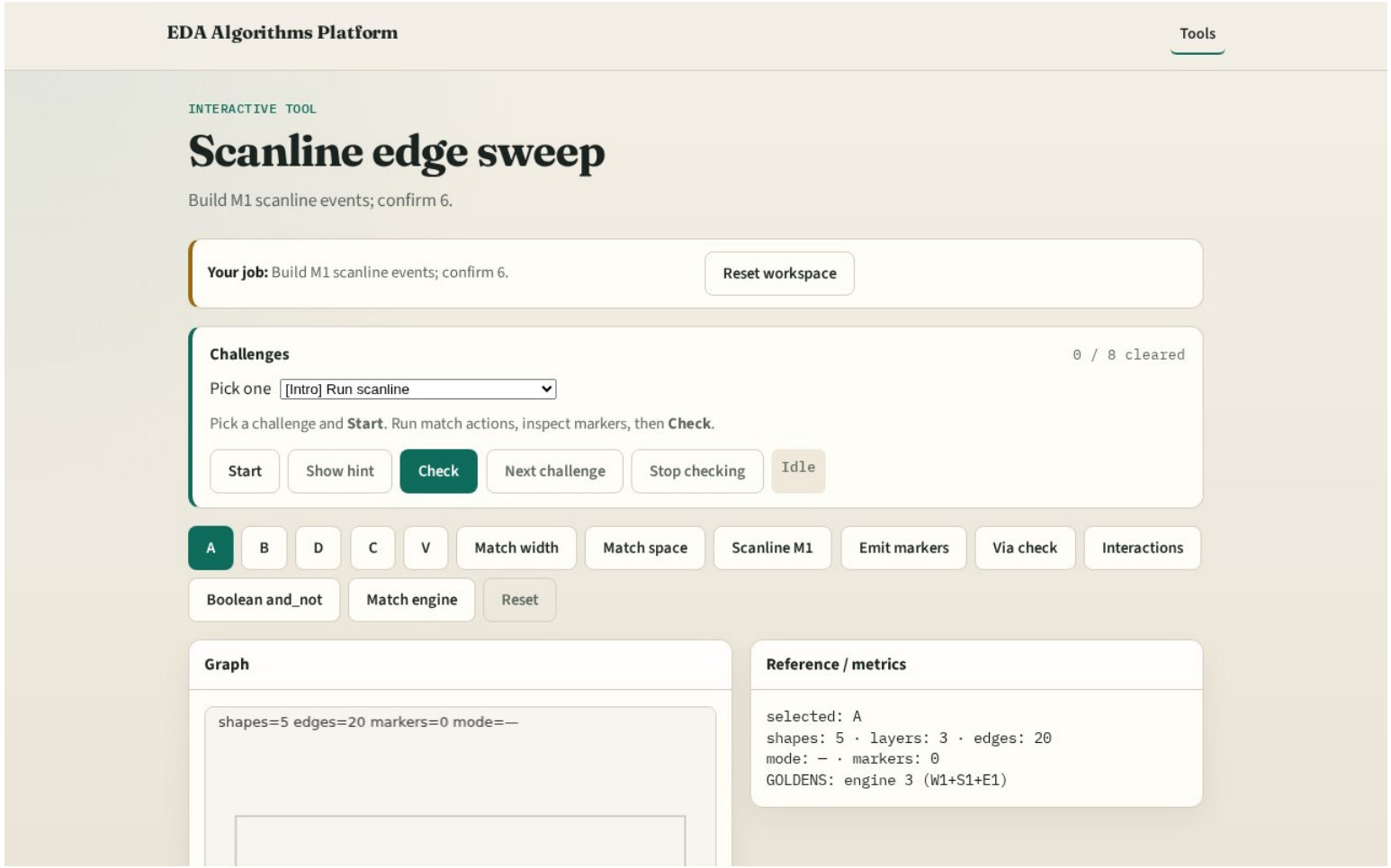
M1 count

Use


SCANLINE SWEEP
Use
Step 5 / 5 · use · module02-01-scanline-sweep
sweep
C/M2
D/M1
A/M1
V/VIA
B/M1
Explanation
Active set between events drives pair checks.
sweep
active

Use

Browser lab track



Browser lab starter

Implement track

- Call scanline_events('M1'); assert len==6

Pitfalls

- Forgetting leave events
- Sorting leaves before enters at same x

Your turn

- Next: marker emission